

WEEK8

Aim:

To develop a simple **Mini Web Application using Node.js and Express.js** with registration, login, dashboard and JSON file storage.

Description:

Develop a **Mini Web Application using Node.js and Express.js** based on a real-world application scenario. The application consists of a **Home/Landing Page, Registration Page, Login Page, and User Dashboard**. Users can register by entering their details, and the registration data is sent to the Express.js backend using a **POST request** and stored in a **JSON file** as a simple database. During login, the entered username and password are checked against the stored user details. If the credentials are valid, the user is allowed to access the dashboard; otherwise, an error message is displayed. The application also provides a **logout option** to return to the login page.

Program Code:

1) app.js:

```
const express = require("express");
const fs = require("fs");
const path = require("path");

const app = express();
const PORT = 3000;

app.use(express.urlencoded({ extended: true }));
app.use(express.json());

app.use(express.static(path.join(__dirname, "public")));

// Home
app.get("/", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "index.html"));
});
```

```

});

// Register page
app.get("/register", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "register.html"));
});

// Register user
app.post("/register", (req, res) => {

  const newUser = {
    name: req.body.name,
    age: req.body.age,
    dob: req.body.dob,
    gender: req.body.gender,
    email: req.body.email,
    mobile: req.body.mobile,
    username: req.body.username,
    password: req.body.password,
    address: req.body.address
  };

  fs.readFile("users.json", "utf8", (err, data) => {

    let users = [];

    if (!err && data.trim() !== "") {
      users = JSON.parse(data);
    }

    users.push(newUser);

    fs.writeFile(
      "users.json",
      JSON.stringify(users, null, 2),
      (err) => {

```

```

        if (err) {
            res.send("Error saving data");
        } else {
            res.send(`
                <h2>Registration Successful!</h2>
                <a href="/login">Go to Login</a>
            `);
        }
    }
});
});

// Login page
app.get("/login", (req, res) => {
    res.sendFile(path.join(__dirname, "public", "login.html"));
});

// Login
app.post("/login", (req, res) => {

    const username = req.body.username;
    const password = req.body.password;

    fs.readFile("users.json", "utf8", (err, data) => {

        if (err) {
            res.send("Error reading data");
            return;
        }

        const users = JSON.parse(data);

        const user = users.find(
            u => u.username === username &&
                u.password === password
        );
    });

```

```

        if (user) {
            res.sendFile(
                path.join(__dirname, "public", "dashboard.html")
            );
        } else {
            res.send(`
                <h2>Invalid Username or Password</h2>
                <a href="/login">Try Again</a>
            `);
        }
    });
});

// Logout
app.get("/logout", (req, res) => {
    res.redirect("/login");
});

// Start server
app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});

```

2)index.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Mini Web Application</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Mini Web Application</h1>
```

```
<nav>
```

```
<a href="/">Home</a>
```

```
<a href="/register">Register</a>
```

```
<a href="/login">Login</a>
```

```
</nav>
```

```
</header>
```

```
<main class="container">
```

```
<h2>Welcome to Our Application</h2>
```

```
<p>
```

```
    Welcome to our Mini Web Application.
```

```
</p>
```

```
<p>
```

```
    This application is developed using
```

HTML, CSS, JavaScript, Node.js and Express.js.

</p>

<p>

New users can register and existing users

can login to access the dashboard.

</p>

<div class="buttons">

Register Now

Login

</div>

</main>

<footer>

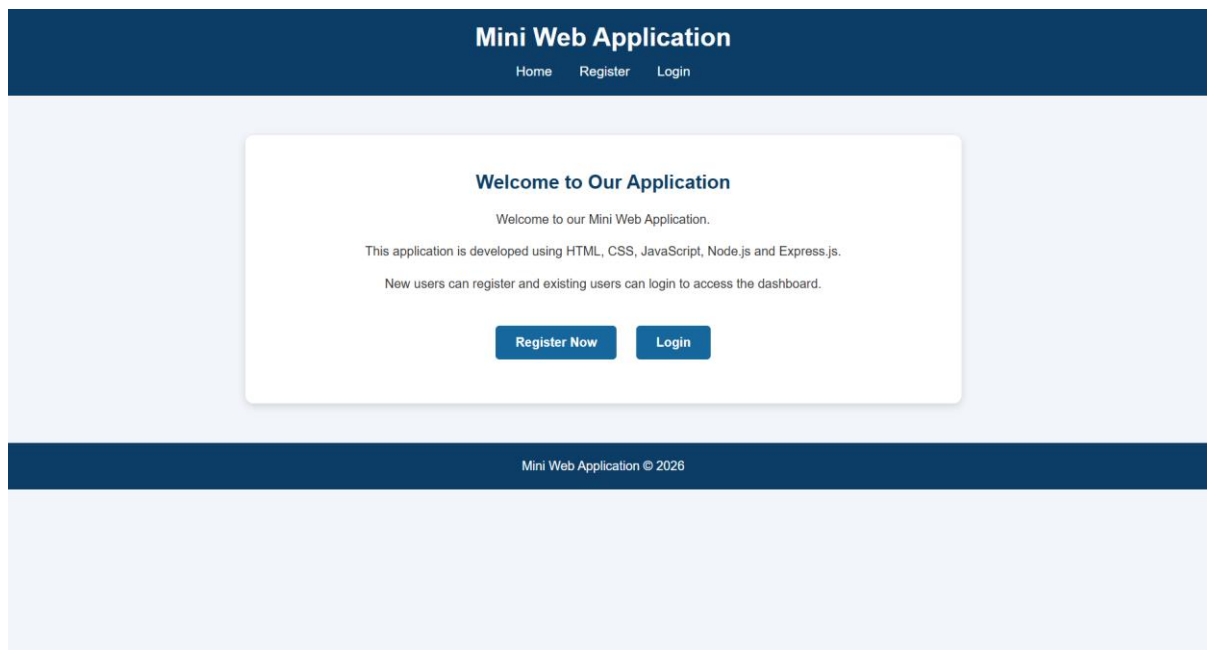
<p>Mini Web Application © 2026</p>

</footer>

</body>

</html>

Output:



3)register.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">
```

```
  <title>User Registration</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <header>
```

<h1>User Registration</h1>

<nav>

Home

Register

Login

</nav>

</header>

<main class="form-container">

<h2>Create Your Account</h2>

<form action="/register" method="POST">

<label>Name:</label>

<input
 type="text"
 name="name"
 placeholder="Enter your name"
 required
>

<label>Age:</label>

<input
 type="number"
 name="age"
 placeholder="Enter your age"
 required
>

<label>Date of Birth:</label>


```
<input
  type="date"
  name="dob"
  required
>
```

```
<label>Gender:</label>
```

```
<select name="gender" required>
```

```
  <option value="">
    Select Gender
  </option>
```

```
  <option value="Male">
    Male
  </option>
```

```
  <option value="Female">
    Female
  </option>
```

```
  <option value="Other">
    Other
  </option>
```

```
</select>
```

```
<label>Email:</label>
```

```
<input
  type="email"
  name="email"
  placeholder="Enter your email"
  required
>
```

```
<label>Mobile Number:</label>
```

```
<input
  type="tel"
  name="mobile"
  placeholder="Enter mobile number"
  required
>
```

```
<label>Username:</label>
```

```
<input
  type="text"
  name="username"
  placeholder="Create username"
  required
>
```

```
<label>Password:</label>
```

```
<input
  type="password"
  name="password"
  placeholder="Create password"
  required
>
```

```
<label>Address:</label>
```

```
<textarea
  name="address"
  rows="4"
  placeholder="Enter your address"
  required></textarea>
```

```
<button type="submit">
  Register
</button>
```

```
</form>
```

```
<p class="bottom-text">
```

Already have an account?

```
<a href="/login">
```

Login here

```
</a>
```

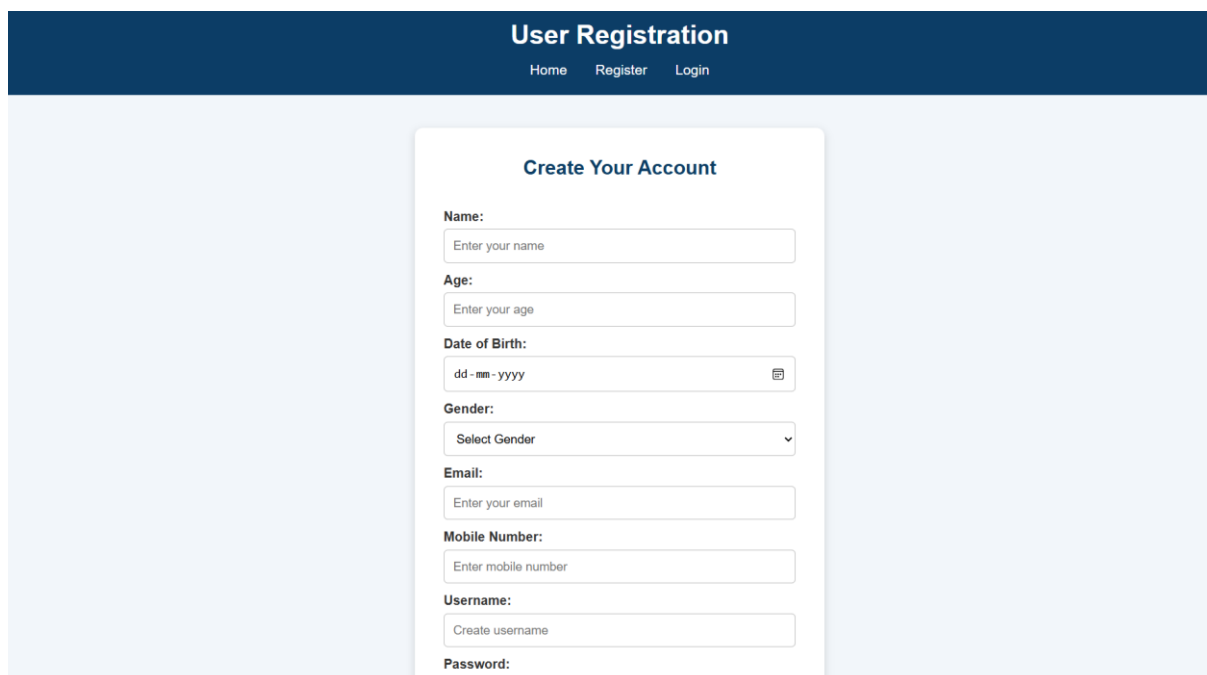
```
</p>
```

```
</main>
```

```
</body>
```

```
</html>
```

Output:



The image shows a web application for user registration. At the top, there is a dark blue header with the text "User Registration" in white. Below the header, there are three links: "Home", "Register", and "Login". The main content area is light blue and contains a white card titled "Create Your Account". The card has several input fields for user information: "Name:" with a placeholder "Enter your name", "Age:" with a placeholder "Enter your age", "Date of Birth:" with a placeholder "dd - mm - yyyy" and a calendar icon, "Gender:" with a dropdown menu showing "Select Gender", "Email:" with a placeholder "Enter your email", "Mobile Number:" with a placeholder "Enter mobile number", "Username:" with a placeholder "Create username", and "Password:" with a placeholder "Create password".

Enter your age

Date of Birth:
dd-mm-yyyy

Gender:
Select Gender

Email:
Enter your email

Mobile Number:
Enter mobile number

Username:
Create username

Password:
Create password

Address:
Enter your address

Register

Already have an account? [Login here](#)

4)login.html:

```
<!DOCTYPE html>
<html>

<head>
  <title>Login</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

<header>
  <h1>User Login</h1>

  <nav>
    <a href="/">Home</a>
    <a href="/register">Register</a>
    <a href="/login">Login</a>
  </nav>
</header>

<div class="form-container">
```

```
<h2>Login</h2>
```

```
<form action="/login" method="POST">
```

```
  <label>Username:</label>
```

```
  <input type="text" name="username" required>
```

```
  <label>Password:</label>
```

```
  <input type="password" name="password" required>
```

```
  <button type="submit">Login</button>
```

```
</form>
```

```
<p>
```

```
  Don't have an account?
```

```
  <a href="/register">Register</a>
```

```
</p>
```

```
</div>
```

```
</body>
```

```
</html>
```

Output:

User Login

[Home](#) [Register](#) [Login](#)

Login to Your Account

Username:

Password:

Login

Don't have an account? [Register here](#)

Style.css:

```
* {  
    box-sizing: border-box;  
}
```

```
body {  
    font-family: Arial;  
    margin: 0;  
    background-color: #f2f6fa;  
}
```

```
header {  
    background-color: #0c3d66;  
    color: white;  
    text-align: center;  
    padding: 20px;  
}
```

```
nav a {  
    color: white;  
    text-decoration: none;  
    margin: 15px;  
}
```

```
.container {  
    width: 80%;  
    margin: 50px auto;  
    padding: 40px;  
    background-color: white;  
    text-align: center;  
    border-radius: 10px;  
}
```

```
.form-container {  
    width: 500px;  
    max-width: 90%;
```

```
    margin: 40px auto;
    padding: 30px;
    background-color: white;
    border-radius: 10px;
}
```

```
form {
    display: flex;
    flex-direction: column;
}
```

```
label {
    margin-top: 10px;
    margin-bottom: 5px;
    font-weight: bold;
}
```

```
input,
select,
textarea {
    padding: 10px;
    border: 1px solid #ccc;
    border-radius: 5px;
}
```

```
button,
.btn {
    margin-top: 20px;
    padding: 12px;
    background-color: #15679d;
    color: white;
    border: none;
    border-radius: 5px;
    text-decoration: none;
    cursor: pointer;
}
```

```
button:hover,  
.btn:hover {  
    background-color: #0c3d66;  
}
```

Users.json:

```
[]
```

Result: Thus, the Mini Web Application using Node.js and Express.js was successfully developed and executed. The application allows users to register, store their details in a JSON file, login using valid credentials, access the dashboard, and logout.